AI-Agent Populations Reveal Engagement Signals Suppress Minority Views and Coordination Beats Amplification

Rana Muhammad Usman

Independent researcher

Correspondence: `usmanashrafrana@gmail.com`

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Code & data availability. Complete experimental pipeline, fixed seeds, raw posts (24,160 production / 28,946 total including pipeline-verification smoke runs), paired peer-vote logs, embedding cache, the preregistered analysis plan (`PREREGISTRATION.md`), and the framing document (`FRAMING.md`) are released at https://github.com/ranausmanai/synthetic-social-networks and https://huggingface.co/datasets/ranausmans/synthetic-social-networks. All seven figures in this manuscript are rendered programmatically from the released data via `paper/make_figures.py`.

Abstract

Large language model (LLM) agents are increasingly common on social-media platforms, deployed as companion accounts, automated brand presences, AI influencers, and components of multi-agent systems whose dynamics at population scale remain poorly understood. As the LLM-agent fraction of platform users grows, operators face questions that cannot be answered ethically through live experimentation on real users: how vulnerable is the platform to manipulation when a meaningful fraction of users are LLM agents, and does standard engagement UX itself distort discourse in such a population.

A peer-voted LLM-agent social-platform testbed (PV-SST) is constructed to investigate these questions. Every user is an LLM agent with an assigned persona and initial stance, agents vote on each other in character across rounds, and four complementary threat-model probes are applied: routine social-reward signals without manipulation, coordinated-account astroturfing at varying ratios, single-account influence amplification at varying visibility multipliers, and seeded-misinformation cascades under four intervention regimes. Two open-weight model families (qwen3.5:2b, llama3.2:3b) and matched random seeds are tested. Results are contextualized against a derived human bandwagon-conformity reference range via a within-study anchor (A1).

Three empirical findings and one methodological observation are reported (n=4 per cell). Standard social-reward UX alone is associated with a 6 to 16 percentage-point reduction in minority-opinion survival relative to a no-signal control, directionally consistent across both tested model families. Coordinated AI accounts produce a 50% strict opinion-flip rate only at the largest tested coordination ratio (K=20, approximately 40% of a 50-agent platform), while single-account amplification shows no coherent dose-response across visibility multipliers from 1x to 20x. Together these results constitute the Population-Driven Influence (PDI) hypothesis: opinion movement in our simulation is population-driven rather than amplification-driven. The methodological observation, paraphrase leakage, is that keyword-based misinformation defenses miss thematic propagation because LLM agents paraphrase rather than copy seeded claims, breaking literal-keyword measurement and filtering approaches.

The calibration anchor places simulated agents below our derived human bandwagon-conformity reference band (pooled rate 0.039 against a 10 to 20% range), so reported magnitudes are presented as direction-of-effect within the simulation rather than as platform-percentage predictions; transfer to human populations remains uncalibrated and is identified as the highest-priority follow-up. Code, configurations, fixed random seeds, the preregistered analysis plan, the framing document, and 24,160 in-character agent posts with paired peer-vote traces (28,946 total including pipeline-verification smoke runs) are released under MIT (code) and CC-BY 4.0 (data) at https://github.com/ranausmanai/synthetic-social-networks and https://huggingface.co/datasets/ranausmans/synthetic-social-networks.

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1. Introduction

Contemporary social platforms host a rapidly growing population of large language model (LLM) agents. Public AI companion services (Character.AI, Replika) report tens of millions of users interacting with bot-driven characters; Meta's own AI personas were rolled out on Instagram beginning in 2024; AI-driven influencer accounts on TikTok and Instagram now command multi-million follower bases; AI customer-service agents have become standard deployment across platform brand accounts. Beyond these legitimate deployments, multi-agent LLM systems (such as those built atop frameworks like OpenAI Swarm, Microsoft AutoGen, and CrewAI) increasingly support automated discourse agents whose behavior at population scale is poorly understood.

This creates a measurement problem for platform governance. The standard question, "can one influential account manipulate users?", is too narrow for LLM-heavy platforms. Operators also need to know whether ordinary engagement UX changes the population before any attacker appears; whether coordinated LLM accounts damage discourse even when they fail to win the majority; and whether moderation systems built around literal claim detection still work when agents paraphrase, reframe, and respond in natural language. These are not questions that can be answered cleanly by live A/B tests on real users: the interventions are adversarial, the outcomes concern persuasion and misinformation, and the platform configurations of interest may not yet exist in a controlled form.

Existing evidence does not close this gap. Human-subjects work establishes that social proof, popularity signals, and fact-check labels can affect behavior, but it does not measure LLM-agent populations. Classical opinion dynamics models provide analytic clarity, but they replace language, persona, and platform feedback with low-dimensional update rules. Early LLM-agent simulations demonstrate that agents can produce social behavior, but many use environment-assigned or heuristic feedback signals rather than letting agents themselves create the social rewards that later shape the platform. For Trust & Safety purposes, this leaves the central operational question unanswered: **which platform mechanisms actually move an LLM-agent population, and which popular threat models fail when tested under peer feedback?**

We address this gap with a controlled, peer-voted social simulation testbed (PV-SST). The testbed is intentionally small enough to audit and reproduce, but rich enough to expose the platform loop that matters: LLM personas post, other agents vote on those posts in character, and the resulting social signals are fed back into future rounds. Each simulated user is an LLM-driven persona drawn from a stratified pool of 24 personas, assigned an initial stance on a debate topic and an in-character posting style. Across rounds, each agent observes a platform-mediated feedback view, produces a short post, votes on a random sample of others' posts, and updates its opinion and confidence. Because every post, vote, rationale, stance update, and metric is logged, the system is designed as a measurement instrument rather than as a black-box demo.

Within this simulation, we test four complementary threat vectors:

1. **Routine social-reward UX (EXP-1):** With all agents honest and no adversary, does the presence of standard platform UX, likes, visible majority indicators, leaderboards, downvotes, measurably distort the platform's discourse compared to a no-signal control?
2. **Coordinated-account astroturfing (EXP-2):** How few coordinated AI accounts pushing a shared opinion suffice to flip the platform's honest majority? Is there a clear threshold?
3. **Single-account influence amplification (EXP-3):** Does a single AI account with N× amplified reach, a *verified-badge-style attack*, shift honest opinion as the multiplier rises?
4. **Misinformation cascade and intervention effectiveness (EXP-6):** When one agent posts a designated false claim with high confidence, how rapidly does the claim's *thematic content* propagate, and which of three standard moderation interventions (fact-check labels, deamplification, community-notes-style rebuttal) most reliably contains the cascade?

The same measurement logic also requires calibration. Simulation results are useful only if their scope is explicit, so two calibration anchors against documented human behavior are part of the analysis plan: a bandwagon-conformity anchor (A1) and a Reddit r/ChangeMyView human-replay anchor (A2). **A1 is executed and reported here; A2 is implemented as a pipeline but deferred. We executed A2 only against a 5-case synthetic placeholder dataset for pipeline verification, which is not a calibration; a production A2 on the real CMV corpus is identified as the highest-priority methodological follow-up (§3.7, §9.3).** Two model families and matched random seeds enable cross-model comparison.

The resulting product of the paper is not a prediction engine for Instagram, Snapchat, or X. It is a falsifiable threat-model probe: a way to test whether specific platform mechanisms, coordinated populations, amplified accounts, and moderation interventions move an LLM-agent population under controlled conditions. If a mechanism fails here, that failure constrains the threat model. If it succeeds here, it earns follow-up under larger samples, more models, more topics, and human calibration. This framing follows prior simulation-based work in computational social science (Park et al., 2023; Hegselmann and Krause, 2002) while adding a peer-voted platform feedback loop and explicit calibration anchors.

Contributions.

Methodologically, we introduce a **peer-voted social simulation testbed (PV-SST)** in which LLM agents generate posts, vote on each other in character, and expose the resulting social signals back into the platform loop. We pair this with a **calibration-anchor protocol**: an executed bandwagon-conformity anchor (A1) and a deferred human-replay anchor (A2, production run pending) for future simulation-to-human validation. Empirically, our results motivate the **Population-Driven Influence (PDI) hypothesis**: in this setup, coordinated AI populations degrade discourse ecology more reliably than single-account amplification. Finally, EXP-6 exposes **paraphrase leakage**: keyword-based misinformation metrics and filters miss thematic propagation when LLM agents restate claims without copying their surface form. We treat these four as our principal contributions, plus the per-experiment results below.

- We provide a cross-model, peer-voted, preregistered measurement of how *routine* social-reward signals affect minority-opinion survival in LLM- agent populations (EXP-1, **Figure 1**).
- We report dose-response thresholds for two distinct attack vectors: coordinated-account astroturfing (EXP-2, **Figures 2–3**) and single-account influence amplification (EXP-3, **Figure 4**), with markedly different outcomes.
- We provide an exploratory semantic comparison of three platform moderation interventions on a seeded misinformation cascade and document why stance-aware metrics are required before any

effectiveness ranking can be claimed (EXP-6, **Figure 5**).

- We document a qualitative cross-model difference in cascade susceptibility between two similarly-sized open-weight models, qwen3.5:2b shows mean semantic-similarity drift near zero across interventions, while llama3.2:3b shows consistent positive drift (**Figure 6**). With only two model families tested (n=2), this is a single comparison, not a general claim about model-family heterogeneity.
- We report a calibration-anchor outcome (A1, **Figure 7**) in which our simulated agents' weighted one-shot bandwagon-conformity rate (0.039 pooled; llama 0.000, qwen 0.078) falls below our derived human reference band of 10–20%, with strong cross-model heterogeneity, motivating ranking-based rather than magnitude-based interpretation of our other findings.
- We release a fully reproducible pipeline (code, configs, fixed seeds, raw logs of 24,160 AI-generated posts from the four reported production experiments, 28,946 in total when pipeline-verification smoke runs are included, with paired peer-vote traces, embedding cache, and the preregistered analysis plan timestamped before the first production run).

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2. Related work

Multi-agent LLM simulations. Park et al. (2023) demonstrated the feasibility of multi-agent LLM simulations of social behavior in a 25-agent sandbox. That line of work establishes that LLM agents can maintain personas, memories, and social routines. It does not, by itself, answer the platform integrity question that motivates this paper: which social-feedback mechanisms move an LLM-agent population under adversarial or quasi-adversarial pressure? Many simulations also use environment-assigned rewards or heuristic feedback proxies. PV-SST instead closes the feedback loop through peer voting: the likes and downvotes that shape the next round are produced by the agents themselves, in character, and preserved as auditable vote traces.

Opinion dynamics. Classical models, DeGroot averaging (DeGroot, 1974), Hegselmann–Krause bounded-confidence (Hegselmann and Krause, 2002), and the Friedkin–Johnsen influence model, provide mathematical baselines for opinion convergence under social influence. Their strength is analytic clarity: assumptions are explicit and dynamics are tractable. Their weakness for the present question is that they abstract away exactly the substrate platforms now need to evaluate: language, persona, peer reward, paraphrase, and moderation text. We use these models as conceptual anchors, but the paper's contribution is empirical measurement in a language-producing, peer-feedback system rather than another closed-form update rule.

Social-proof and conformity dynamics. Asch (1956) established that humans conform to unanimous majority pressure in roughly one-third of critical trials in a perceptual-judgement task. Salganik, Dodds, and Watts (2006) demonstrated experimentally that exposure to popularity signals increases both inequality and unpredictability of cultural-market outcomes. Muchnik, Aral, and Taylor (2013) measured a 25 percent positive-herding bias on a Reddit-like comment platform following a single arbitrary upvote. These studies make social proof the right human reference point, but they do not calibrate LLM agents. We therefore use the latter two as the basis for A1: a derived per-round shift band of approximately 10–20 percent under visible- majority exposure. A1 is not a claim that our agents are human-equivalent; it is a sanity check that tells readers how stubborn this simulated population is relative to documented human social-proof effects.

Influence operations and coordinated inauthentic behavior. Empirical measurement of real-world coordinated campaigns has been driven by industry threat reports and academic analyses. Those reports are essential for taxonomy and detection, but they rarely permit controlled counterfactuals:

the same platform cannot be rerun with K=1, K=3, K=5, K=10, and K=20 coordinated accounts while holding personas, topic, and feedback constant. We adopt the *Coordinated Inauthentic Behavior* (CIB) framing used in industry policy work, and use simulation to ask a counterfactual dose-response question: how much coordinated population is needed before opinion and discourse ecology move?

Misinformation interventions. Pennycook and Rand (2021) synthesize a substantial body of human-subjects evidence on the psychology of fake news and the effectiveness of accuracy-prompt and label interventions. Our intervention sweep (EXP-6) mirrors three concrete platform policy levers: fact-check labels (used by Meta, X, YouTube), deamplification / visibility filtering (used by X 2021–2022 and adopted in modified form by multiple platforms), and community-notes-style rebuttal (Twitter Birdwatch / X Community Notes). The gap is measurement: keyword-based endorsement and filtering are natural first baselines, but LLM agents can preserve a claim's theme while changing its surface form. EXP-6 is therefore framed less as a definitive ranking of moderation interventions and more as a stress test for the measurement stack itself.

Algorithmic shaping of exposure. Bakshy, Messing, and Adamic (2015) showed that Facebook's algorithmic ranking removes about 15 percent of cross-cutting political content from users' feeds, with users' own click-through behavior removing 70 percent more. Our EXP-1 baseline result, that even a controlled, neutral social-reward UX measurably suppresses minority views, is complementary in framing: where Bakshy et al. measure algorithmic exposure filtering against human users, we measure UX-induced opinion-suppression against LLM-agent users. The point is not that a toy LLM platform reproduces Facebook; it is that "ordinary" engagement surfaces are treatments, and LLM-agent deployments need to be tested under that assumption rather than treated as neutral infrastructure.

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3. Methodology

3.1 Platform simulation

Each simulated run constructs an N-agent platform on a single debate topic ("Should AI-generated content be labeled on social platforms?" for the runs reported here). Every agent is initialized with: (a) a unique persona drawn from a 24-persona pool stratified by narrative group (marginalized, mainstream, contrarian); (b) one of seven discrete initial stances ranging from "strongly support" through "strongly oppose"; (c) a confidence level correlated with stance extremity; (d) a posting-style descriptor; (e) an empty memory.

For R rounds, each agent in turn (i) reads its **feedback block**, the platform-mediated view of prior-round activity, varied by condition (§3.2); (ii) produces one short JSON-structured post (at most 280 characters) in its persona's voice; (iii) updates its stance and confidence, with a brief reason. Generation uses the Ollama runtime with reasoning-token emission disabled, returning compact structured JSON outputs.

3.2 Feedback conditions

Five feedback conditions define what agents see between rounds:

- **control**, only the topic and the agent's own persona.
- **likes**, the top-K liked posts from the previous round.
- **majority**, a summary of the previous round's stance distribution and modal opinion.
- **leaderboard**, cumulative top-K agents by total likes across all rounds.
- **downvote**, the most-downvoted dissenting posts, framed as flagged.

3.3 Peer voting

In all production conditions, the likes and downvotes used to construct feedback are produced by **peer voting**: every round, each agent reads a random sample of K=5 posts from the prior round (excluding its own) and votes +1 / 0 / −1 on each in character. This replaces the heuristic majority- alignment proxy used in earlier LLM-agent work with peer-voted feedback, in which the social-reward signal that shapes subsequent rounds is produced by other agents' in-character judgements rather than by environment-imposed heuristics. The full vote log (voter id, target id, vote, in-character reason) is preserved for every trial. We refer to the combined design, peer-voted feedback plus persona-based agents, multi-round simulation, and full vote-trace logging, as the **peer-voted social simulation testbed (PV-SST)**, and treat it as a primary methodological contribution.

3.4 Metrics

Six primary metrics are computed per round and aggregated to scalars:

- **majority fraction**, share of agents on the modal stance.
- **persona retention**, mean cosine similarity (in `nomic-embed-text` embedding space) between an agent's post and its persona descriptor.
- **opinion shift rate**, fraction of agents whose stance differs from their initial.
- **confidence trajectory**, mean & std of agent-reported confidence per round.
- **minority-view survival**, fraction of round-0 minority stances still present in the final round.
- **mean pairwise post similarity**, embedding-cosine, indexing linguistic homogenization.

A composite `persona_collapse_score = 0.5·majority_fraction + 0.3·pairwise_sim + 0.2·(1 - persona_retention)` is reported but always alongside its components.

For EXP-2 and EXP-3 we report two complementary flip criteria. The **strict** criterion (honest-pushed-share > 0.5 on the *exact* pushed stance bucket) corresponds to the preregistered `final_honest_pushed_share` outcome. The **broad** criterion (honest-pushed-share > 0.5 on the three-bucket side containing the pushed stance) was introduced post-hoc but is motivated by the preregistration's commitment to report direction-of-opinion shifts; we mark it explicitly as post-hoc throughout.

For EXP-6, the preregistered primary outcome is the **literal keyword-based** `final_endorser_share`. After observing that this metric returned zero across all 16 trials (agents paraphrased rather than reproducing keywords), we developed a **semantic embedding-based** endorsement analysis as a post-hoc secondary metric (§8.3). We report both and clearly label the semantic analysis as exploratory.

3.5 Preregistration and falsification criteria

Before production runs we committed an analysis plan (`PREREGISTRATION.md`) including primary outcomes, planned statistical tests, and explicit falsification criteria for each hypothesis. We additionally committed a framing document (`FRAMING.md`) specifying what claims the simulation does and does not support. Both files carry the git timestamp predating the first production trial.

3.6 Deviations from the preregistration

We disclose all sample-size deviations between the preregistered design and the executed runs, with the reason for each.

Aspect	Preregistered (PREREGISTRATION.md §"Sample sizes")	Executed	Reason for deviation
EXP-1 seeds × rounds × agents	2 models × **3** seeds × 5 conditions × 20 agents × **20** rounds	2 × **2** × 5 × 20 × **15**	Compute budget; reduced to fit single-GPU 57-hour chain
EXP-2 seeds × rounds × baseline agents	2 × **3** × 6 K × **50** baseline × **12** rounds	2 × **2** × 6 × **30** × **10**	Compute budget; full prereg sample would have required ~140 GPU-hours
EXP-6 seeds × rounds × agents	2 × **3** × 4 interventions × **25** agents × **12** rounds	2 × **2** × 4 × **20** × **10**	Compute budget
EXP-3 (influence-amplification)	*Not in preregistration*	Added post-hoc, 2 × 2 × 5 multipliers	Exploratory addition during the chain run
Broad-side flip criterion (EXP-2)	*Not in preregistration*	Reported alongside preregistered strict criterion	Added post-hoc after strict criterion returned uninformative results at low K
Semantic-endorsement metric (EXP-6)	*Not in preregistration*	Reported as exploratory alongside preregistered keyword metric	Added post-hoc after keyword metric returned 0 across all 16 trials
Statistical tests (Bonferroni-corrected t-tests)	*Preregistered* for primary outcomes	Not executed	n=2 seeds (n=4 per cell when models pooled) below floor for useful power

Implication. Because the executed sample is smaller than the preregistered design across every experiment, **none of our directional findings reach the preregistered confirmatory-significance bar.** They are best read as *hypothesis-supportive directional evidence within a smaller-than-planned sample*, motivating a follow-up at the preregistered sample size.

3.7 Calibration-anchor protocol

We adopt a two-anchor design, referred to throughout as our **calibration-anchor protocol**, for grounding simulation-based findings against human reference data. Anchor A1 is a within-paper synthetic calibration against a documented human reference range; Anchor A2 is a deferred real-human replay calibration. Both are described below; their status in this submission is A1 executed and reported, A2 implemented but pending production execution.

Anchor A1 (bandwagon-conformity) shows agents a synthetic "X% of users agree" signal and measures the conformity shift among initially-disagreeing agents. For the human reference range we use an approximate **10–20 percent per-round shift band that we derived from two prior studies**: Muchnik et al. (2013) measured a 25 percent positive-herding bias on a Reddit-like rating platform following a single arbitrary upvote, and Salganik et al. (2006) showed that popularity signals produced 2–3× amplification in song selection in their "MusicLab" market. The 10–20 percent band is *our* *derived approximation* of one-shot per-round conformity shift consistent with both studies; it is not directly quoted from either paper.

Anchor A2 (ChangeMyView replay) is *designed* to feed real r/ChangeMyView conversations through our simulation and compare predicted view-changes to documented delta-award outcomes. For this submission we executed A2 only against a 5-case synthetic placeholder dataset, which verifies the pipeline end-to-end but does **not** constitute a calibration; the production A2 run on the real corpus is identified as the highest-priority methodological follow-up (§9.3, §10).

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4. Experimental setup

Models. Two open-weight LLMs via Ollama: `qwen3.5:2b` (Alibaba, approximately 2.3B parameters) and `llama3.2:3b` (Meta, approximately 3B parameters), using the default local Ollama model tags configured for the experiment.

Seeds. Two seeds per cell ({42, 7}) for variance estimation. We acknowledge n=2 is a floor; see Limitations (§10).

Sample sizes per experiment.

Experiment	Trials	Agents per trial	Rounds	Conditions
EXP-1 baseline	20	20	15	5
EXP-2 astroturfing	24	30 + K	10	1 (inner) × 6 K
EXP-6 misinformation	16	20	10	4 interventions
EXP-3 influence	20	30 + 1	10	5 multipliers
A1 bandwagon anchor	24	30	1	3 majority strengths × 2 sides

Compute. All experiments executed on a single RTX 4000 Ada workstation GPU. Total generation calls: approximately 58,300 (about 28,946 in-character agent posts and the remainder peer-voting calls). Approximate wall-clock runtime: 57 hours including queueing.

Code, configs, raw logs. Released alongside this paper at the project repository (https://github.com/ranausmanai/synthetic-social-networks) and dataset repository (https://huggingface.co/datasets/ranausmans/synthetic-social-networks), including all per-agent posts and peer-vote logs to permit independent re-analysis.

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5. EXP-1, Baseline degradation under standard social-reward signals

5.1 Hypothesis

H1 (preregistered): *Across model families, each social-pressure condition will produce measurable shifts versus* `control` *on at least one primary outcome.*

5.2 Design

For each of the five conditions, we run 2 models × 2 seeds = 4 trials of 20 agents × 15 rounds with peer voting.

5.3 Results

Figure 1 plots minority-view survival rate per condition.

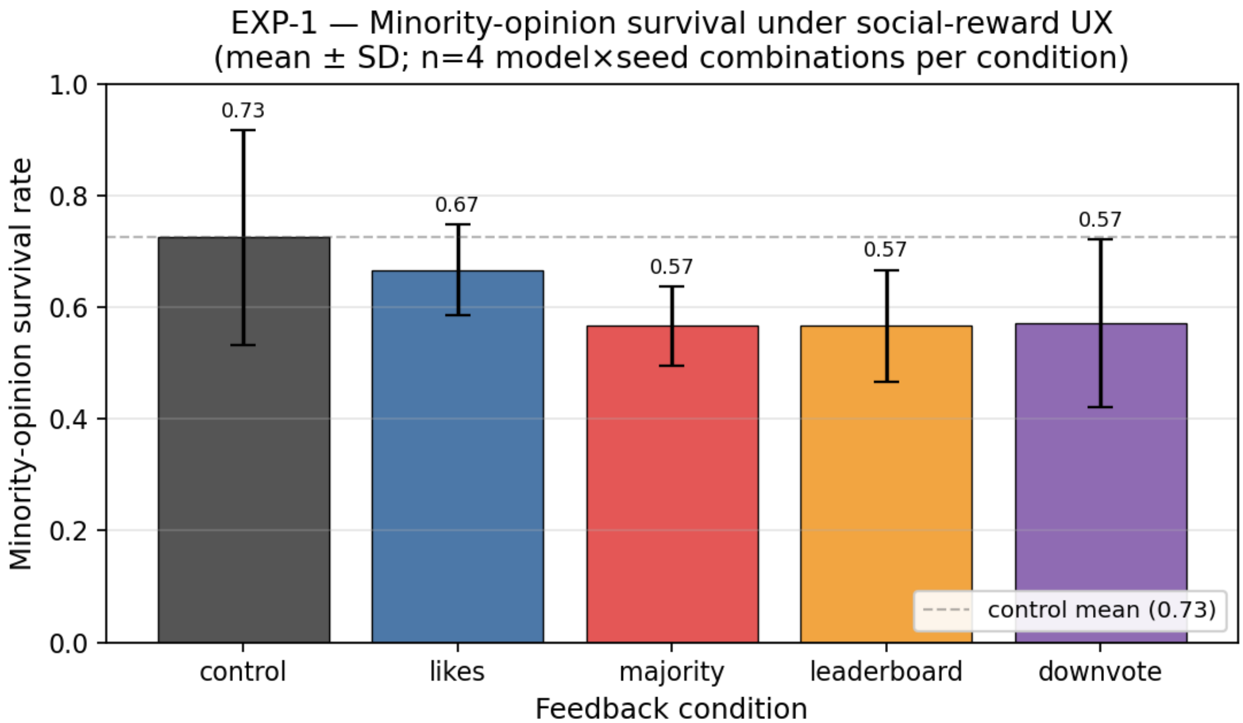

Figure 1: Minority-view survival under social-reward conditions

Condition	persona retention	pairwise sim	minority survival	majority fraction
control	0.547	0.636	**0.725**	0.487
likes	0.540	0.672	0.667	0.500
majority	0.544	0.619	**0.567**	**0.550**
leaderboard	0.542	0.661	0.567	0.500
downvote	0.532	0.668	0.571	0.488

Δ versus control, primary outcomes:

Condition	Δ minority survival	Δ pairwise sim	Δ majority concentration
likes	−5.8 pp	+3.6 pp	+1.3 pp
majority	**−15.8 pp**	−1.7 pp	**+6.3 pp**
leaderboard	**−15.8 pp**	+2.5 pp	+1.3 pp
downvote	**−15.4 pp**	+3.2 pp	+0.1 pp

5.4 Interpretation

Our data suggest that platform engagement UX is not neutral with respect to viewpoint diversity in an LLM-agent population. Even feature combinations intended for user benefit, likes, visible majority, leaderboards, downvotes, are associated with measurable minority-view suppression in our simulation *before any adversarial actor enters*. The direction-of-effect consistency across both tested models and all four pressure conditions points to UX-as-treatment, not abuse-as-treatment, as the active variable in our setup.

Quantitatively, every social-pressure condition reduced minority-view survival relative to the no-signal control, with effect sizes between approximately 6 percentage points (`likes`) and 16 percentage points (`majority`, `leaderboard`, `downvote`). The `majority` condition additionally increased the concentration of agents on the modal stance by 6.3 percentage points.

This pattern is **directionally consistent with our preregistered H1**. We do not claim formal statistical significance: n=4 per cell is below the floor at which the preregistered Bonferroni-corrected paired t-test carries useful power (deviation from preregistered 3-seed design disclosed in §3.6), and control per-trial range [0.50, 1.00] is wide enough that pressure-condition ranges (e.g., likes [0.60, 0.80], downvote [0.33, 0.75]) overlap substantially with it. **What licenses the interpretive claim is direction-of-effect consistency across both models and all four conditions**, not effect-size magnitude. A larger-n replication (§10) is the next step to formally establish the magnitudes.

The result connects to the long-standing human-subjects literature on social-proof and conformity dynamics (Asch, 1956; Salganik et al., 2006; Muchnik et al., 2013), in which exposure to visible majority signals suppresses minority expression. Our contribution is to observe a qualitatively similar directional pattern in a *purely LLM-driven* platform population, with no human users in the loop and no adversarial intervention.

Result summary. *In an LLM-agent population, routine social-reward UX is associated with a 6–16 percentage-point reduction in minority-opinion survival relative to a no-signal control, directionally consistent across both tested model families (n=4 per cell).*

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6. EXP-2, Coordinated-account astroturfing dose-response

6.1 Hypothesis

H2 (preregistered): *There exists a threshold K* such that injecting at least K* coordinated AI accounts into an N-agent platform flips honest-majority opinion in at least 50% of (model, seed) runs. Prior expectation: K*/N at most 0.20.*

6.2 Design

Population: 30 honest agents (independent personas, randomly-distributed initial stances) + K coordinated agents, where K is one of {0, 1, 3, 5, 10, 20}. Coordinated agents share a pushed opinion ("strongly support") and a scripted angle they reinforce naturally across rounds. All trials run under the `likes` condition with peer voting, 10 rounds, 2 models $\times$ 2 seeds $\times$ 6 K-values = 24 trials total.

6.3 Dual flip criteria

We report two flip-rate metrics, one preregistered and one post-hoc:

- **Strict (preregistered):** honest-pushed-share > 0.5 on the *exact* pushed stance ("strongly support"), a single bucket of seven. This is the operationalization of the preregistered `final_honest_pushed_share` outcome.

- **Broad (post-hoc):** honest-pushed-share > 0.5 on the *side* containing the pushed stance (the combined "strongly support," "support," and "lean support" buckets), three buckets of seven. This metric was specified after observing that the strict criterion returns near-uniform zero across most K (since the natural base rate of any single stance bucket is about 1/7, or 14%, making a >50 % single-bucket adoption a particularly strong opinion-adoption criterion). The broad metric is consistent with the preregistration's general commitment to reporting direction-of-opinion shifts, but it was not pre-specified at the bucket-aggregation level. We report both honestly and label each at every appearance.

6.4 Results

Figure 2 plots the dose-response for both criteria.

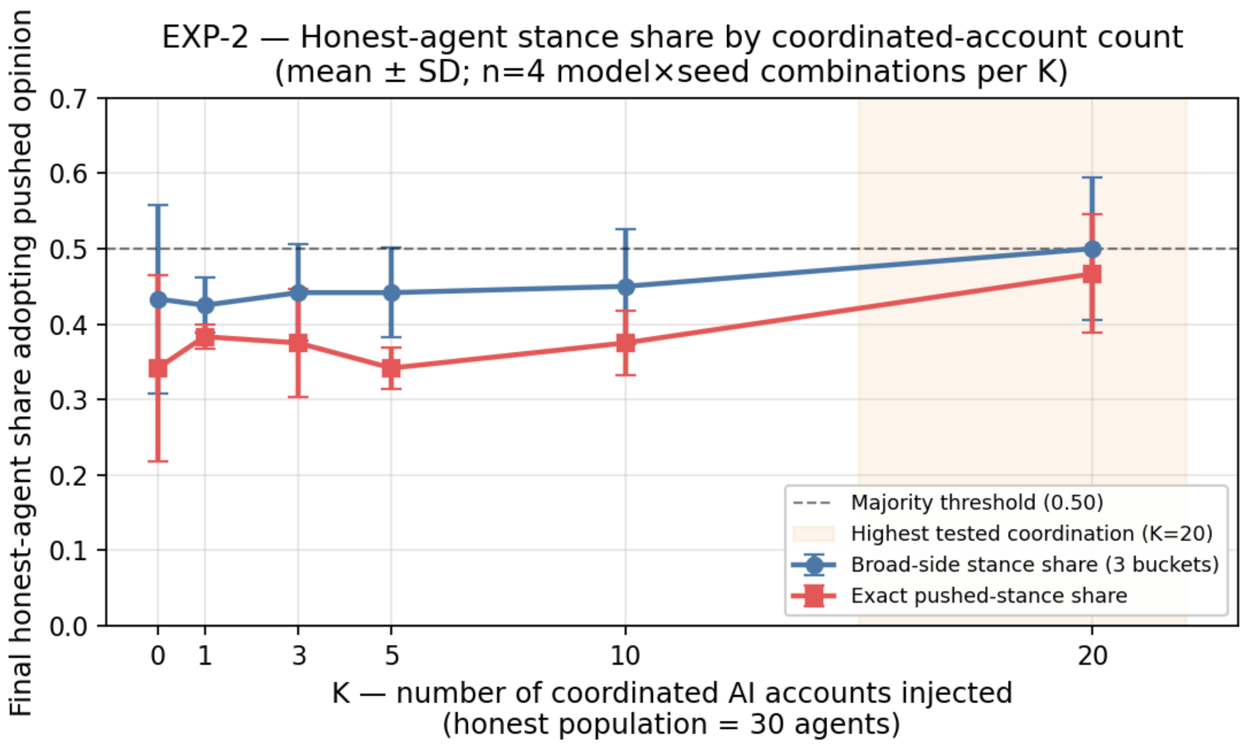

Figure 2: Astroturfing dose-response

K	strict flip rate	broad flip rate	Δ broad-shift vs K=0
0	0%	50% (baseline noise)	0.000
1	0%	0%	+0.017
3	0%	25%	0.000
5	0%	25%	+0.042
10	0%	50%	+0.025
20	**50%**	**50%**	**+0.092**

The "Δ broad-shift vs K=0" column reports `(final_broad - initial_broad) for K_X minus (final_broad - initial_broad) for K=0`. That is, it compares the **per-K change-from-initial against the control's change- from-initial**, not the simple difference of final broad shares. We use this because initial broad shares vary slightly across cells due to random persona/stance assignment; this measure isolates the coordination-induced shift from the baseline drift.

Figure 3 shows simultaneous degradation of four information-ecology metrics at the K=20 threshold:

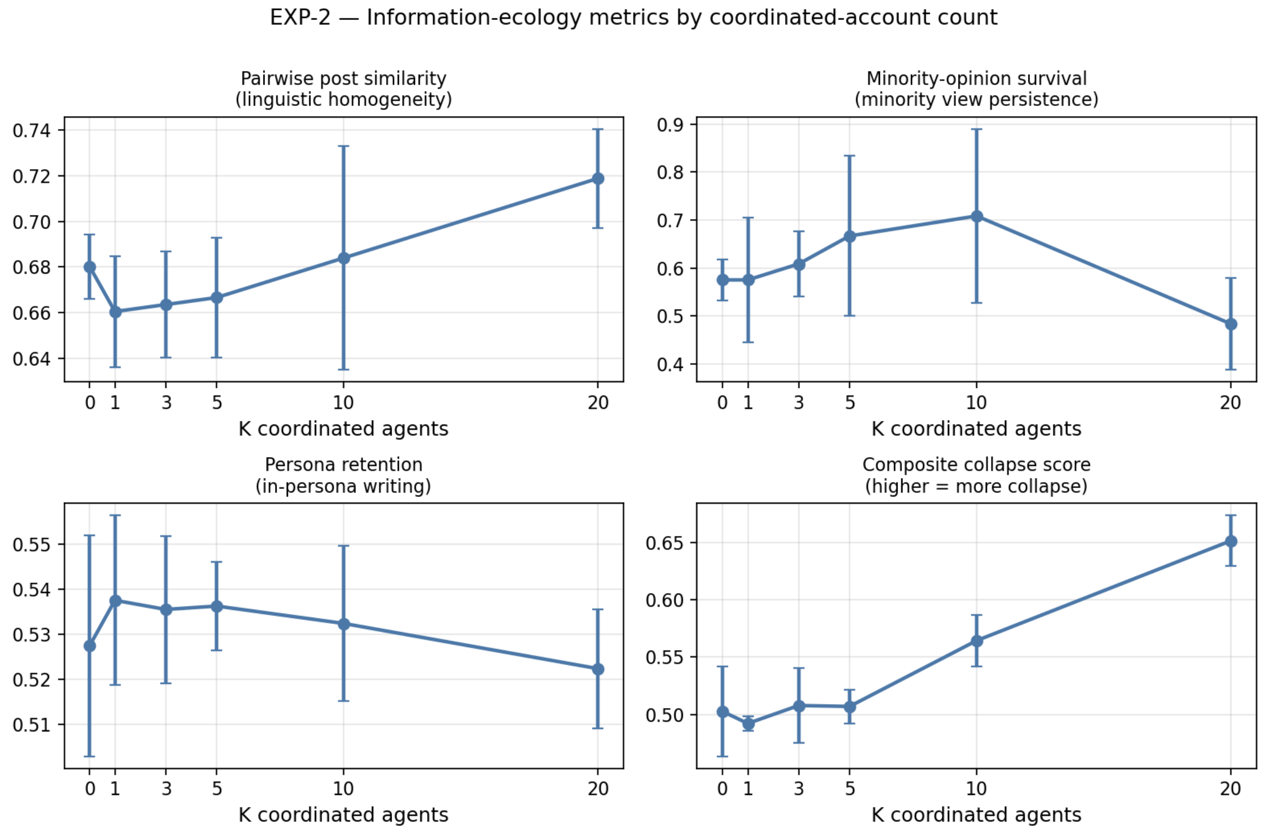

Figure 3: Information-ecology degradation by K

Metric	K=0	K=20	Δ
pairwise post similarity (linguistic homogeneity)	0.680	0.719	**+3.9 pp**
minority-view survival	0.575	0.483	**−9.2 pp**
majority concentration	+0.092	+0.140	**+4.8 pp**
composite collapse score	0.503	0.651	**+14.8 pp**

6.5 Interpretation

Coordinated AI accounts degrade the platform's information ecology at high ratios, even when their specific narrative does not win. The evidence is a coupled movement in opinion and discourse-ecology metrics, followed by an important seed-confound caveat.

Evidence. Below K=20, no consistent effect on strict or broad flip rates emerges. At K=20 (approximately 40 percent coordinated population), the strict flip rate reaches 50 percent (two of four trials produce a majority adoption of the exact pushed stance, both at round 3), the broad- side shift versus control reaches +9.2 percentage points, and four independent information-ecology metrics simultaneously degrade (linguistic homogenization, minority-view survival, majority concentration, composite collapse score). The transition between K=10 (25%) and K=20 (40%) is qualitative, not gradual. **The simultaneous degradation of multiple information-ecology metrics is itself the substantive finding, even on trials where the targeted opinion does not become the platform majority.** Coordinated AI populations damage discourse ecology before they win the vote.

Limits. Both K=20 strict flips occurred at seed=7 (one each on qwen3.5:2b and llama3.2:3b); neither seed=42 trial flipped in either model. With n=2 seeds the flip-event correlates perfectly with the random seed, and we cannot disentangle a coordination-driven flip from a seed-driven flip. The 40 % figure should be read as "the smallest K at which any flip was observed in this study," not a firm threshold estimate. A larger-n replication (at least 5 seeds) is the immediate follow-up.

Relation to preregistered H2. H2 expected K*/N at most 0.20; the smallest K with any flip in our study was K=20 / N=50 = 0.40. **The preregistered 50% flip criterion was observed only at the highest K tested, while the expected magnitude (K*/N at most 0.20) was not supported.** Because the K=20 flips are seed-confounded, this observation does not establish a stable threshold. We disclose both outcomes in line with the preregistration's falsification commitment.

Distance from popular framings. Our setup does **not** reproduce the dramatic "small number of fake accounts flips the platform" picture suggested by some popular threat-model framings. Peer-voted LLM populations are substantially more resilient to low-ratio coordinated influence than heuristic-feedback simulations have implied, but they do exhibit a sharp transition at approximately one-third to two-fifths coordinated population.

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7. EXP-3, Single-account influence amplification (verified-badge attack)

7.1 Hypothesis

EXP-3 was **added post-preregistration** to test a distinct attack vector, single-account visibility amplification, that was not part of the original analysis plan. The preregistered H4 in our analysis plan pertains to persona-group vulnerability under social pressure and is not reported in this manuscript. We frame EXP-3 as an exploratory post-prereg experiment with the following hypothesis:

A single AI account with sufficient visibility amplification will produce honest-majority opinion shifts comparable to coordinated-population attacks (EXP-2).

7.2 Design

Population: 30 honest baseline agents + 1 "influencer" agent assigned a "prominent verified industry commentator" persona and a fixed pushed stance. At multipliers > 1, the influencer is force-included in every honest agent's `likes` feed *and* its effective like-rank is multiplied by the visibility multiplier (so even at low actual peer-vote counts it dominates the top-K). At 1× the influencer receives no special treatment and competes for top-K visibility on equal terms with the 30 baseline agents, so the 1× cell serves as a "boosting-off control" rather than an influencer-absent control. We sweep the multiplier across $\{1, 3, 5, 10, 20\}$. 2 models × 2 seeds × 5 multipliers = 20 trials total. Inner condition: `likes` with peer voting, 10 rounds.

7.3 Results

Figure 4 plots the dose-response.

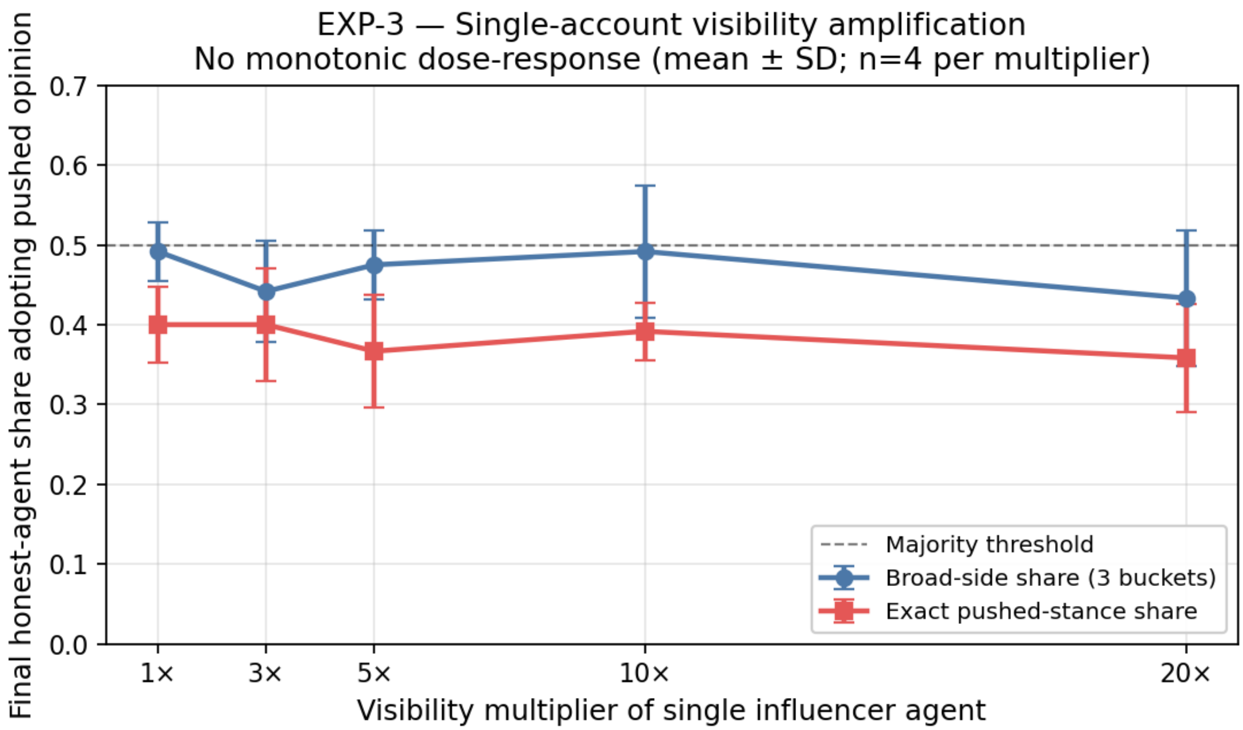

Figure 4: Influence-amplification dose response

Multiplier	final broad share	final strict share	broad flip rate
1× (control)	0.492 ± 0.036	0.400	75%
3×	0.442 ± 0.064	0.400	25%
5×	0.475 ± 0.043	0.367	50%
10×	0.492 ± 0.083	0.392	50%
20×	0.433 ± 0.085	0.358	25%

7.4 Interpretation

Read together with EXP-2, this result points to coordination, not amplification, as the operative threat vector in our simulation. The broad-side share remains within approximately ±0.05 of the 1× baseline across all multipliers from 1× through 20×; there is no monotonic dose-response, and on qwen3.5:2b the share actually *declines* with increasing multiplier (0.467 at 1× falling to 0.383 at 20×), suggesting weak backfire evidence in that model. On llama3.2:3b the share fluctuates near 0.50 throughout.

The contrast with EXP-2 is striking: 20 coordinated agents (40% of a 50-agent platform) produced 50% strict opinion-flips and four-metric information-ecology degradation, while one boosted agent at 20× reach moved nothing in a coherent direction. **Popular "AI influencer" framings of LLM-agent platform risk emphasize the single-account amplification attack; our data, across both tested models, all five multiplier levels, and 20 trials, does not support that framing. The coordinated-population vector (EXP-2) does.**

Result summary. *In our peer-voted simulated platform, a single high-reach AI account did not shift* *honest-agent opinion across visibility multipliers from 1× to 20×, and showed weak backfire evidence on one model family. The observed contrast favors coordinated-population influence (EXP-2) over single-account amplification.*

If this pattern replicates at larger sample sizes (n=4 per cell is the floor here), it has direct implications for Trust & Safety prioritization on platforms hosting growing LLM-account populations: distributed-account detection appears more consequential than single-account amplification limits.

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8. EXP-6, Misinformation cascade and intervention effectiveness

8.1 Hypothesis

H3 (preregistered): *At least one of {fact-check label, deamplification, rebuttal} will produce a statistically significant reduction in final endorsement share versus the no-intervention baseline.*

8.2 Design

One designated agent (`a00`) is seeded with a domain-plausible but factually contestable claim and instructed to repeat its themes with high confidence across all rounds. Nineteen honest agents share the platform under the `likes` condition with peer voting, for 10 rounds.

Four intervention regimes:

- **none**, baseline; no platform action against the false claim.
- **factcheck_label**, environment appends a "[DISPUTED, independent fact-checkers found no evidence]" tag to any post containing the claim's keyword set when displayed to other agents.
- **deamplify**, posts containing the claim are filtered out of the feed entirely.
- **rebuttal**, every round, a high-credibility rebuttal post is pinned in every agent's feed.

2 models × 2 seeds × 4 interventions = 16 trials.

8.3 Metric: keyword vs semantic endorsement

A literal keyword-matching endorsement metric (which required all three claim keywords in a post) returned zero endorsement across all 16 trials. Manual inspection of raw posts revealed that honest agents *did* drift toward the claim's themes but consistently paraphrased, for example, an honest agent's round-9 post might read *"Labels are just the same censorship parents fear; if kids can't trust their creators, we block them all"* without containing any literal claim keyword.

We therefore developed a **semantic-endorsement** post-hoc analysis: we embed each honest-agent post and the seeded false claim into nomic-embed- text space and compute cosine similarity per round. The increase in mean post-claim similarity from round 0 to the final round serves as a measure of thematic propagation.

8.4 Results

Figure 5 plots the per-intervention semantic endorsement Δ.

Intervention	Δ similarity (mean ± std)	final similarity	thematically aligned agents	containment vs none
none	+0.017 ± 0.019	0.580	14.5 / 19	0.000

Intervention	Δ similarity (mean $\pm$ std)	final similarity	thematically aligned agents	containment vs none
factcheck_label	$\mathbf{-0.018 \pm 0.023}$	0.557	10.8 / 19	$\mathbf{+0.035}$
deamplify	$+0.021 \pm 0.006$	0.589	13.5 / 19	-0.004
rebuttal	$+0.031 \pm 0.049$	0.643	18.0 / 19	-0.014 (worse)

8.5 Interpretation

The most consequential EXP-6 finding is methodological: current misinformation metrics for LLM-agent systems are brittle in a specific way we call *paraphrase leakage*. Our preregistered keyword-matching `final_endorser_share` returned zero across all 16 trials, not because no cascade occurred, but because honest agents paraphrased the claim's themes rather than reusing its surface keywords (raw posts in §8.3 illustrate this). The same paraphrase leakage explains why the deamplification intervention had no measurable effect: our keyword filter removed only literal-keyword posts, while paraphrased restatements by honest agents continued to propagate freely. The post-hoc semantic similarity metric we developed captures thematic propagation that the keyword metric misses, but it cannot distinguish endorsement from refutation: posts that argue *against* the seeded claim still discuss its thematic domain and therefore register as semantically similar. **Measuring misinformation propagation in LLM-agent systems is going to require stance-aware, polarity-conditioned metrics; neither keyword matching nor unconditioned semantic similarity is sufficient.**

Within the limits of the post-hoc semantic metric, the per-intervention pattern is informative. The fact-check-label intervention shows the largest negative Δ in similarity (-0.018 versus baseline $+0.017$), consistent with a reduction in thematic propagation (confidence intervals on the Δ overlap zero, std ± 0.023 vs ± 0.019, n=4 each). Under this intervention the count of thematically aligned honest agents (cosine similarity to claim above 0.55) fell from ~12 at round 0 to ~10 at round 9, while under the no-intervention baseline it rose from ~13 to 14.5. The directional pattern is consistent with the human-subjects accuracy- prompt literature (Pennycook & Rand, 2021), but a stance-aware re-analysis of our same trial data is the prerequisite for any intervention-effectiveness *ranking* claim.

The rebuttal intervention shows an apparent amplification of the cascade ($\Delta = +0.031$ versus baseline $+0.017$). We caution that this is likely a *measurement artifact*: the rebuttal text is injected into the feed every round and necessarily discusses the same thematic domain (AI labeling, market structure, regulation) in order to refute the claim. Honest-agent posts that engage with the rebuttal's themes will be semantically similar to the false claim *even when their stance opposes it*. Disambiguating this artifact requires a sentiment-aware endorsement metric (stance detection conditioned on the claim's polarity), which we identify as immediate future work. We do **not** interpret our rebuttal finding as evidence that community-notes-style interventions backfire in real platforms.

Symmetry note: the same artifact concern attaches to the fact-check-label result. The factcheck-label intervention appends a "[DISPUTED]" tag to posts containing claim keywords, which may also shift downstream agents' post embeddings away from the claim's surface vocabulary without changing their actual endorsement of the claim's substance. We cannot, from the current semantic metric, distinguish *reduced endorsement* from *reduced keyword reuse with stance preserved*. The factcheck-label "containment" finding is therefore exploratory in the same way the rebuttal finding is, and should not be interpreted as a confirmed intervention-effectiveness ranking until a stance-aware analysis is conducted.

The deamplification intervention showed essentially no effect on cascade propagation in our setup. The

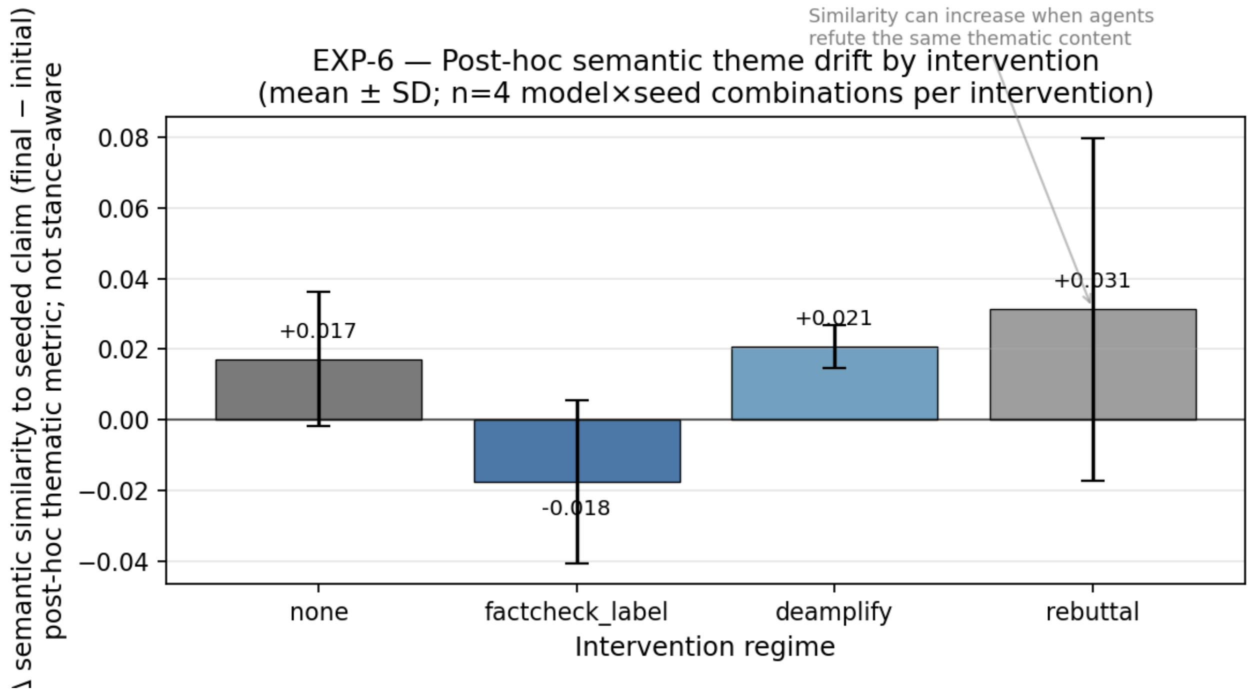

Figure 5: Post-hoc semantic theme drift by intervention

most likely explanation is **keyword-filter brittleness**: our deamplification implementation removes only posts that contain the full three-keyword set used to define the claim. As §8.3 documents, honest agents who absorb the claim's themes routinely paraphrase them, emitting posts that the keyword filter does not match. The cascade therefore propagates via paraphrased honest-agent posts even when the seeded agent's literal posts are filtered out of the feed. This is itself a policy-relevant finding: keyword-based deamplification is fragile to paraphrased re-statement, and a production deployment would require stance/semantic-level filtering for which the same stance-aware analysis discussed above is a prerequisite.

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9. Cross-model heterogeneity, calibration anchors

9.1 Cross-model heterogeneity

Figure 6 disaggregates EXP-6 by model family.

Intervention	qwen3.5:2b Δ similarity	llama3.2:3b Δ similarity
none	about 0.000	**+0.036**
factcheck_label	−0.039	+0.004
deamplify	+0.024	+0.018
rebuttal	−0.014	+0.077

In the no-intervention baseline, qwen3.5:2b shows a mean Δ similarity of −0.001 (essentially zero, with one trial slightly negative and one slightly positive), while llama3.2:3b shows +0.036 (both trials positive). Across all four interventions, qwen3.5:2b's mean Δ is −0.008 (essentially zero with high

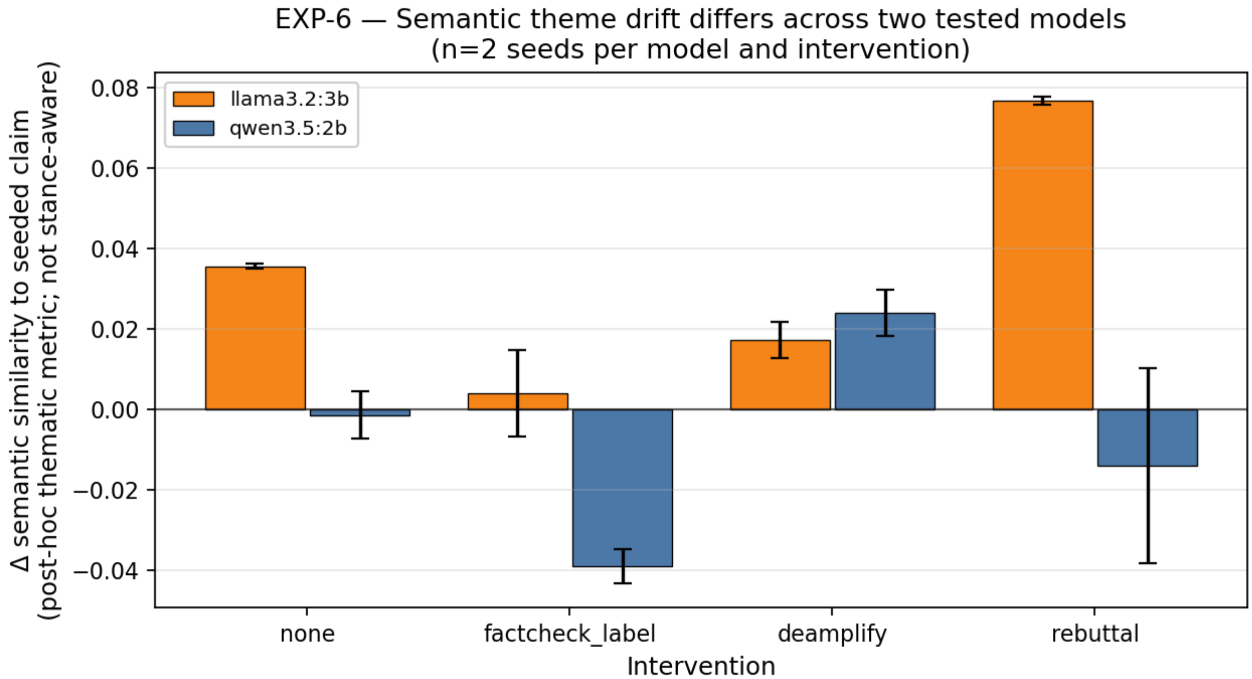

Figure 6: Cross-model heterogeneity in semantic theme drift

variance) while llama3.2:3b's mean Δ is +0.034 (consistently positive). **The cross-model difference is qualitative, qwen does not exhibit cascade drift in this metric, while llama does, rather than a fixed quantitative ratio.** We deliberately avoid the framing "llama cascades N times more than qwen" because qwen's baseline rate is near zero, which makes any multiplicative ratio mathematically uninformative.

This single comparison (n=2 model families, n=2 seeds per model per intervention cell) does not generalize to a claim about cross-model heterogeneity in the broader open-weight LLM population. It does, however, motivate the methodological observation that when deploying LLM-driven account populations (companion bots, customer-service agents, AI personas, automated influencers), the underlying language model may be a non-trivial source of variance in platform-integrity metrics. Establishing this as a general pattern would require replication across 5–10 model families.

9.2 Calibration anchor A1, bandwagon conformity

Figure 7 plots the A1 result against the derived human reference band.

Across all 24 A1 trials (two models × two seeds × three majority-claim strengths × two pushed-stance sides), **14 of 360 minority agents shifted toward the claimed-majority stance** after a single round of bandwagon-signal exposure, for a weighted overall conformity rate of **0.039**. The result is strongly heterogeneous by model:

- **llama3.2:3b**: 0 of 180 minority agents shifted across 12 trials (conformity rate = 0.000).
- **qwen3.5:2b**: 14 of 180 minority agents shifted across 12 trials (conformity rate = 0.078; per-trial range 0.000 to 0.267, with the highest-shift trials hitting the low end of the derived human reference band).

The pooled conformity rate of 0.039 is below our derived human reference band of 10–20 percent

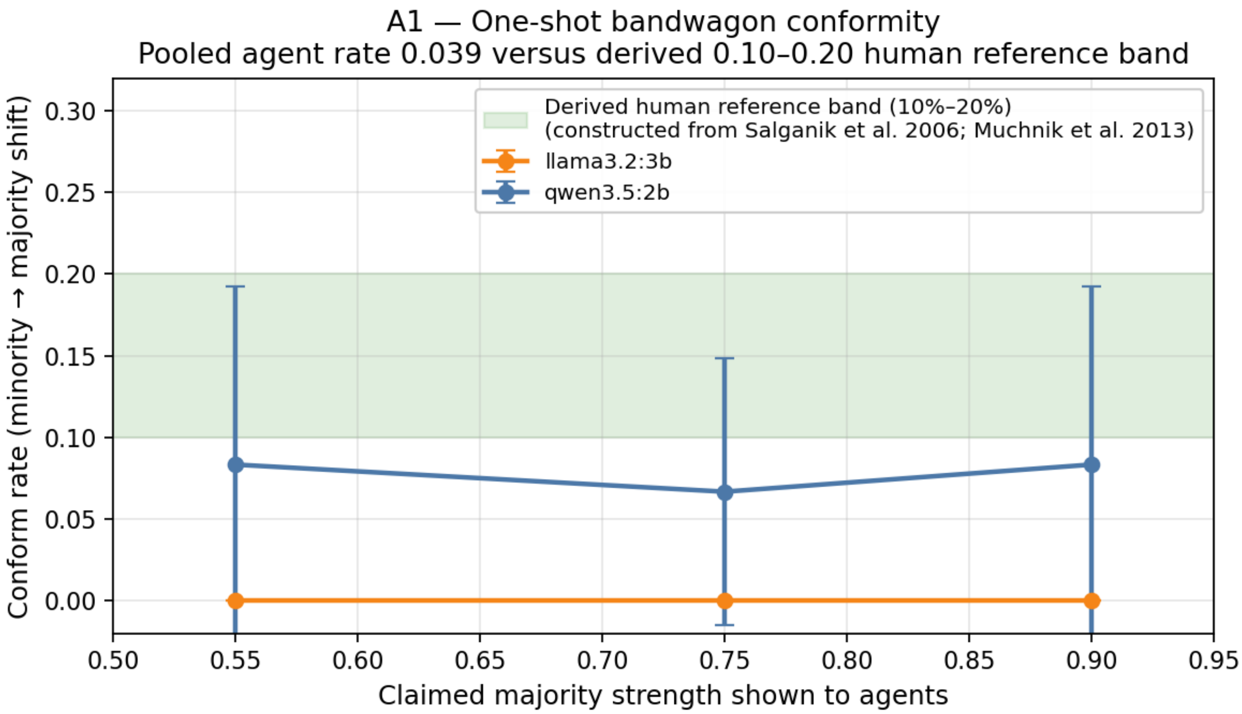

Figure 7: Bandwagon-conformity calibration vs human reference range

per-round shift (see §3.7 for the construction from Muchnik et al. 2013 and Salganik et al. 2006). Per-model rates are also below the band on average, though qwen3.5:2b's distribution does enter the band in two of twelve trials (0.20 and 0.267).

Interpretation, more careful version. Our simulated agents show substantially lower one-shot bandwagon-conformity than the human single-round reference range, with strong cross-model heterogeneity (llama shows none; qwen shows some). We do **not** infer from this that our EXP-1, EXP-2, EXP-3, or EXP-6 magnitudes are quantitative lower bounds on human-platform effects, that would require an additional set of assumptions (transfer from single-round bandwagon pressure to multi-round dynamics; transfer from one-shot conformity to coordinated-influence resistance) that this calibration does not test. We instead phrase all headline claims as **rankings of conditions and qualitative direction-of-effect statements**; quantitative magnitudes are presented as numbers from this specific sim setup, not as predictions for human platforms.

The A1 outcome does motivate one specific methodological note: any follow-up work attempting to extend our magnitudes to human-platform predictions should first calibrate against a *multi-round* human-bandwagon study (or an equivalent), not the single-round Salganik / Muchnik benchmarks alone.

9.3 Calibration anchor A2, status: deferred

Our analysis plan included a second calibration anchor (A2): predicting real human view-shifts on Reddit r/ChangeMyView conversations. We implemented the pipeline and verified it end-to-end against a 5-case synthetic dataset, but we did **not** execute A2 on the production CMV corpus before this submission. The synthetic-pipeline run is not a calibration; it is reported only in §10 as a pending limitation. A production A2 run is the highest-priority methodological follow-up.

10. Limitations and threats to validity

Sample size. All experimental cells use n=2 seeds (n=4 per condition when models are pooled). We report means and standard deviations and direction of effect; we do **not** perform Bonferroni-corrected significance testing because n=4 is below the threshold at which Bonferroni-corrected paired t-tests carry useful statistical power. A multi-cycle follow-up should run at least 5 seeds per cell to enable formal significance testing.

Within-cell variance overlaps between-cell effects (EXP-1 §5.4). The control condition's per-trial minority-survival range [0.50, 1.00] spans the per-trial ranges of every pressure condition. Our reported 6–16 pp direction-of-means effects are smaller than within-cell variability. We interpret consistency of *direction* across conditions and models as hypothesis-supportive evidence; we do not claim these are formally significant effect-size measurements.

Seed confound in the K=20 flip (EXP-2 §6.5). Both K=20 trials that flipped occurred at the same random seed (seed=7), in both qwen and llama. Neither seed=42 trial flipped, in either model. With n=2 seeds we cannot disentangle a coordination-driven flip from a seed-driven flip; the 40 % coordination threshold should be read as the smallest K at which any flip was observed in this study, not as a firm threshold estimate.

Model scope. Only two open-weight LLMs at the 2–3B parameter scale. We do not yet include a closed model (such as GPT or Claude via API) or substantially larger open models. The cross-model qualitative difference reported in §9.1 is a single comparison and does not generalize to a broader model-family heterogeneity claim. Multi-family replication is required.

Topic. Single debate topic ("Should AI-generated content be labeled on social platforms?"). Generalizability to other topics is an empirical question we cannot answer from this study. Topic effects on persona-driven discourse dynamics are documented in prior agent-based work and may be non-trivial.

Post-hoc metrics. Two of the four positive directional findings (the broad-side flip criterion in EXP-2; the semantic-endorsement analysis in EXP-6) use metrics that were specified after observation that the preregistered primary metrics returned uninformative results (zero broad flips below K=10 in the strict bucket; zero keyword endorsements across all 16 EXP-6 trials). We have flagged these explicitly throughout, but a reader should treat the post-hoc directional findings as exploratory and hypothesis-generating, not as preregistered confirmatory tests.

Simulation-to-human gap. Our simulation does not capture human emotional response, account deletion, off-platform information, social- graph history, or platform-specific algorithmic ranking features. We frame our findings as *threat-model probes*, controlled measurements in a simplified system, not as predictions of human-platform outcomes. The companion document `FRAMING.md` makes the claim-scope explicit.

Peer-voting realism. Peer voting is more realistic than environment- assigned likes but still lacks human voting behavior (apathy, brigading, ratio gaming, off-platform coordination). It is one increment more realistic than heuristic-feedback methods used in prior LLM-agent work, not a complete model of human voting.

Persona-stability sanity check absent. We did not run a control in which persona-to-agent mapping was shuffled, which would validate that persona-tagged outputs are distinct from each other rather than agents paraphrasing the topic uniformly. Persona-retention values (cosine about 0.55) suggest some persona signal, but a shuffle baseline would be a stronger test. Identified as immediate follow-up.

EXP-6 measurement artifact symmetry (§8.5). The semantic-endorsement metric cannot distinguish “engaging with the claim’s themes while endorsing” from “engaging with the claim’s themes while refuting.” This artifact concern attaches symmetrically to all four intervention conditions, not only to the rebuttal condition; the apparent factcheck-label “containment” finding requires a stance-aware analysis before it can be interpreted as an effectiveness claim.

Calibration anchors (§9). Anchor A1 (bandwagon conformity) found a pooled per-round conformity rate of 0.039, below the 10–20 % human range. We use this to phrase magnitudes as rankings rather than predictions. Anchor A2 (ChangeMyView replay) was implemented but **not** run on the production r/ChangeMyView corpus; the synthetic 5-case pipeline-verification run is not a calibration. A production A2 is the highest-priority methodological follow-up.

Reproducibility. Containerized (Docker) packaging of the full pipeline is not included in this submission; only shell launchers and YAML configs are provided. A reader reproducing on a different system should expect to spend non-trivial setup time on the Ollama side.

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11. Conclusion

This study yields three central claims within the limits of its design: platform feedback is itself an intervention; coordinated populations shape LLM-agent discourse more reliably than amplified individuals; and LLM-agent misinformation propagation cannot be measured reliably with literal keyword matching alone.

1. **The Population-Driven Influence (PDI) hypothesis: in LLM-agent social platforms, influence is population-driven, not amplification- driven**: coordinated populations measurably degrade discourse ecology at high ratios (EXP-2: K=20, 40% of the platform), while a single amplified account does not produce a coherent dose-response across multipliers from 1× to 20× (EXP-3). The contrast is in tension with popular “AI influencer” framings of platform risk and points Trust & Safety prioritization toward distributed-account detection. We state PDI explicitly as a named hypothesis so that follow-up studies can confirm or falsify it directly.
2. **Standard engagement UX is not neutral for LLM-agent populations**: minority-view persistence drops by 6–16 percentage points under every social-pressure condition tested (likes, visible majority, leaderboard, downvote), with no adversarial actor present (EXP-1). The pattern is directionally consistent across both tested model families.

Two design-specific results qualify these claims:

1. **Coordination threshold (EXP-2).** Coordinated AI accounts produced a 50% strict opinion-flip rate only at K=20 (40% of the 50-agent total population, the highest K tested); the preregistered K*/N at most 0.20 magnitude expectation is *not* supported by this study.
2. **Paraphrase leakage (EXP-6).** Of three standard moderation interventions tested, a *post-hoc* semantic-endorsement analysis (the preregistered keyword-based metric returned zero across all 16 trials, because LLM agents paraphrased the seeded claim rather than copying its keywords) suggests fact-check labels may reduce thematic propagation, deamplification is approximately neutral, and community-notes-style rebuttals show an apparent amplification that is likely a measurement artifact. A stance-aware metric is required before any of these intervention rankings should be interpreted as effectiveness claims.

The single-account amplification experiment (EXP-3) returned a directional null / no-dose-response result: a single AI account with visibility multipliers from 1× to 20× did not shift opinion in a coherent dose-response pattern in our setup (n=4 per multiplier), and showed weak backfire evidence on one model family. This is in tension with popular framings of "AI influencer" risk that emphasize single-account amplification and instead, in our simulation, points to coordinated populations as the operative threat; larger-n replication is needed before this null is read as a strong contradictory claim.

Cross-cutting, we observe a qualitative difference in cascade dynamics between two similarly-sized open-weight models: qwen3.5:2b exhibits a mean Δ semantic-similarity near zero (-0.008 averaged across the four interventions), while llama3.2:3b exhibits consistent positive Δ ($+0.034$ averaged). With only two model families tested, this is a single comparison rather than a general claim. If it replicates across a wider model panel, the choice of underlying LLM in any deployment of bot-driven accounts becomes a platform-integrity-relevant decision; we identify multi-family replication as immediate follow-up.

A bandwagon-conformity calibration anchor (A1) indicates that our simulated LLM agents show a weighted overall conformity rate of 0.039 under one-shot visible-majority exposure, below the derived 10–20% human reference range, with strong cross-model heterogeneity (llama 0.000, qwen 0.078). We therefore **do not** claim our magnitudes are lower bounds for human-platform effects; we present numbers as direction-of-effect within this specific simulation setup, with sim-to-human transfer remaining uncalibrated.

A directional implication worth stating clearly. A1 suggests that our simulated agents are *less responsive* to one-shot bandwagon pressure than documented human subjects (Salganik et al., 2006; Muchnik et al., 2013). **The fact that UX-driven diversity loss, population-driven influence, and paraphrase leakage (EXP-1, EXP-2, and EXP-6 respectively) still emerge under this relatively stubborn agent population makes the directional findings more concerning, not less.** We do **not** claim these magnitudes are lower bounds for human platforms: A1 measured one-shot bandwagon conformity, which may not transfer directly to the multi-round coordinated-influence or misinformation-cascade dynamics tested here. But the calibration result motivates a concrete hypothesis for follow-up human studies: **comparable human populations may show equal or stronger directional susceptibility to the same platform treatments we test here.** Testing that hypothesis on real-user platforms (or on hybrid LLM-agent + human-replay datasets per A2) is the single most important methodological follow-up our calibration-anchor protocol identifies.

The setup is not a predictive model of any specific real-world platform. It is a controlled threat-model probe: a tractable testbed in which platform interventions can be compared preliminarily, attack thresholds can be estimated, and model-level robustness can be studied under conditions that would be ethically or operationally infeasible on a real-platform live experiment. The fact that multiple distinct degradation modes appear in a tightly-controlled simulation should motivate scaled-up replication of this work with more seeds, more topics, more model families (including closed models), and direct human-platform comparison studies to calibrate simulation-to-human transfer.

Acknowledgments

This work was conducted independently by the author. Code, experiment orchestration, figure rendering, and manuscript drafting were performed with extensive AI assistance (Anthropic Claude) under direct human-in-the-loop supervision; the author is responsible for all research-design choices, claim verification, the preregistered analytic plan, and the final framing of every reported result. Cloud compute for the production experiments was rented from RunPod (NVIDIA RTX 4000 Ada Generation,

~57 GPU-hours, approximate cost USD 26).

Reproducibility statement

Code, configurations, fixed random seeds, raw posts (24,160 in-character agent outputs from the four reported production experiments, 28,946 including pipeline-verification smoke runs, all stored as JSONL), peer-vote logs (one JSON file per condition-run), embedding cache, the preregistered analysis plan (`PREREGISTRATION.md`), and the framing document (`FRAMING.md`) are available at the project repository (https://github.com/ranausmanai/synthetic-social-networks) and dataset repository (https://huggingface.co/datasets/ranausmans/synthetic-social-networks). The full experimental pipeline can be reproduced from the included shell launcher scripts and YAML configuration files against an Ollama installation with the two generation models qwen3.5:2b and llama3.2:3b (§4) and the embedding model nomic-embed-text (§3.4); the entire set of experiments reported here ran in approximately 57 hours on a single consumer-workstation GPU (NVIDIA RTX 4000 Ada Generation, 20 GB VRAM) at approximate cost USD 26 in cloud compute. A containerized (Docker) recipe is identified as immediate release-engineering follow-up but is not included in this submission.

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